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Low profile wire connector

Abstract

The present disclosure provides descriptions of configurations for wire connectors used to splice low voltage electrical conductors or to splice high voltage electrical conductors. The wire connectors may be coupled to form a wire connector assembly. Each wire connector includes a base, a cover, a first cover coupling assembly and a second first cover coupling assembly. The base has an upper wall and a lower wall and can be divided into a first end portion, a second end portion and a central portion between the first end portion and the second end portion. The cover has a main cover member, a first movable cover member movably connected to a first side of the main cover member and a second movable cover member movably connected to a second side of the main cover member. The main cover member includes a plurality of contact chambers, where each contact chamber includes at least one electrical contact positioned within a cavity of the contact chamber. The first cover coupling assembly is associated with the first end portion of the base and the first movable cover member of the cover. The second cover coupling assembly is associated with the second end portion of the base and the second movable cover member of the cover. The cover assemblies are used to releasably couple the movable cover members to the base.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present disclosure is based on and claims benefit from U.S. Provisional Patent Application Ser. No. 63/291,100 filed on Dec. 17, 2021 entitled “Low Profile Wire Connector” the contents of which are incorporated herein in their entirety by reference.

BACKGROUND

Field

(1) The present disclosure relates generally to wire connectors, and more particularly to low-profile wire connectors used to connect low voltage wires and line voltage wires.

Description of the Related Art

(2) More and more buildings, homes, etc. are being built utilizing smart building technology. Such smart building technology includes but is not limited to Light-Emitting-Diode (LED) lighting, fluorescent lighting including dimming systems as well as other power, control and signal circuits. To control smart building technology, generally low voltage control/signal wiring (sometimes referred to generally as low voltage wiring or control conductors) and line voltage wiring (sometimes referred to generally as power conductors) are run throughout the building, home, etc.

(3) Current electrical code requires a divider to exist in a junction box and connectors to electrically isolate the low voltage wiring (e.g., about 42.4V AC max or about 30V DC max) from the line voltage (e.g., from about 120 VAC to about 277 VAC) wiring. Such a divider is generally a thin plastic or metal wall. The wire runs for control/signal circuits and power circuits are also required to be in separate sheathing, e.g., metal-sheathing.

(4) Recently, approved cables have been introduced to the market that have a double insulated set of low voltage wires and a double insulated set of line voltage wires. An example of such a cable is an MC-PCS cable. The set of low voltage wires are individually insulated by insulating jackets and a second insulating sleeve or jacket surrounds the set. Likewise, the set of line voltage wires are individually insulated by insulating jackets and a second insulating sleeve or jacket surrounds the set. Both the low voltage wires and the line voltage wires run in the same protective sheathing, e.g., metal-sheathing, to save cost. Separator tape may also be provided between the wires and the inside of the sheathing. The low voltage wires typically carry control signals such as control signals used for LED lighting systems or other smart building technology. The line voltage wires typically carry power.

(5) The low voltage wires are generally smaller in diameter than the line voltage wires. For example, the line voltage wires are generally 10 to 14 AWG wires while the low voltage wires are generally 18 to 24 AWG wires. The line voltage wires and the low voltage wires may be solid or stranded depending on a particular application.

(6) Since low voltage wires are run in the same conduit as line voltage wires, there is a concern that electricity from the line voltage wires may jump to the low voltage wires, which may cause hazardous conditions, such as fire or equipment damage. The National Electrical Code (NEC) that governs the separation of cables of this type is NEC Section 725.136. This Section of the NEC specifies either, 1) a separation of 0.25 inches between the low voltage wires and line voltage wires, or 2) the insulation of the low voltage wires have the same insulation factor as those used for line voltage wires, e.g., 30 mil jacket over the low voltage wires, which is the same cumulative thickness as those used for line voltage wires.

(7) Low voltage wires inside the same outer sheathing as the line voltage wires satisfy the NEC by using the same 30 mil insulation jacket to achieve the same insulation thickness as the line voltage wires. However, inside an electrical junction box where the low voltage wires are typically removed from the sheathing and a portion stripped and spliced with other low-voltage wires, the NEC requirement to maintain the integrity of the cumulative insulation thickness cannot be satisfied with the same insulator type because some of the insulation jacket has been stripped off. Thus, there is a need for a termination connection device that satisfies the insulation thickness or spacing requirement for these types of jacketed sets of insulated control conductors from the high voltage power conductors of the unsheathed portions of these cables.

SUMMARY

(8) The present disclosure provides exemplary embodiments for low profile wire connectors and

wire connector assemblies used for splicing low voltage electrical conductors or high voltage electrical conductors. In one exemplary embodiment, a wire connector includes a base, a cover, first cover coupling assembly and second cover coupling assembly. The base includes a first end portion, a second end portion and a central portion between the first and second portions. The base also includes an upper wall and a lower wall. The cover includes a main cover member, a first movable cover member movably connected to a first side of the main cover member and a second movable cover member movably connected to a second side of the main cover member. The cover is attached to the base such that the main cover member is associated with the central portion of the base, the first movable cover member is associated with the first end portion of the base and the second movable cover member is associated with the second end portion of the base. The main cover member of the cover includes a plurality of contact chambers. Each contact chamber includes at least one electrical contact positioned within a cavity of the contact chamber, a first passage to the first end portion of the base and a second passage to the second end portion of the base. The first cover coupling assembly is associated with the first end portion of the base and the first movable cover member of the cover. The first cover coupling assembly is used to releasably couple the first movable cover member to the first end portion of the base. The second cover coupling assembly is associated with the second end portion of the base and the second movable cover member of the cover. The second cover coupling assembly is used to releasably couple the second movable cover member to the first end portion of the base.
(9) An exemplary of a wire connector assembly includes a first wire connector described above and a second wire connector described above, where the base of the first wire connector is attached to the base of the second wire connector.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The figures depict embodiments for purposes of illustration only. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures illustrated herein may be employed without departing from the principles described herein, wherein:
(2) FIG. **1** is a perspective view of an exemplary embodiment of a low-profile wire connector according to the present disclosure;
(3) FIG. **2** is a perspective view of the wire connector of FIG. **1**, illustrating a cover of the low-profile wire connector in an open position;
(4) FIG. **2**A is a cross-sectional view of a portion of the wire connector of FIG. **2** taken from detail **2**A of FIG. **2**, and illustrating an exemplary embodiment of a cover coupling assembly according to the present disclosure;
(5) FIG. **3** is an exploded top perspective view of the wire connector of FIG. **1**, illustrating electrical cables and components of the low profile wire connector including a cover, a base and electrical contacts;
(6) FIG. **4** is an exploded bottom perspective view of the wire connector of FIG. **1**, illustrating electrical cables and components of the low profile wire connector including a cover, a base and electrical contacts positioned in contact chambers of the cover;
(7) FIG. **5** is a bottom perspective view of the cover of the wire connectors of FIG. **4**, illustrating that the electrical contacts removably held within the cover.
(8) FIG. **6** is an enlarged perspective view of an exemplary embodiment of an electrical contact of FIG. **5** taken from detail **6** of FIG. **5**;
(9) FIG. **7** is an enlarged top plan view of a portion of the cover of FIG. **4** taken from detail **7** of FIG. **4**, and illustrating the contact chamber of the cover holding the electrical contact which is

electrically connecting two electrical conductors;
(10) FIG. **8** is a perspective view of an exemplary embodiment of a wire connector assembly according to the present disclosure, illustrating two of the wire connectors of FIG. **1** coupled together;
(11) FIG. **9** is an exploded perspective view of the wire connector assembly of FIG. **8**, illustrating cables and components of the two wire connectors including covers, bases and electrical contacts;
(12) FIG. **10** is a perspective view of the wire connector assembly FIG. **9**, illustrating the two wire connectors just prior to coupling;
(13) FIG. **11** is a cross-sectional view of the wire connector assembly of FIG. **10** taken from detail **11** of FIG. **10**, and illustrating an exemplary embodiment of the two bases positioned to be coupled to each other;
(14) FIG. **12** is a cross-sectional view of the wire connector assembly of FIG. **11**, illustrating the two bases just prior to being coupled to each other;
(15) FIG. **13** is a cross-sectional view of the wire connector assembly of FIG. **10** taken from detail **13** of FIG. **10**, and illustrating an exemplary embodiment of the two bases positioned to be coupled to each other;
(16) FIG. **14** is a cross-sectional view of the wire connector assembly of FIG. **11**, illustrating the two bases just prior to being coupled to each other;
(17) FIG. **15** is a perspective view of another exemplary embodiment of a low-profile wire connector assembly according to the present disclosure;
(18) FIG. **16** is an exploded perspective view of the low-profile wire connector of FIG. **15**, illustrating cables and components of the low-profile wire connector including multiple covers and a common base;
(19) FIG. **17** is a perspective view of an exemplary embodiment of a low-profile wire connector according to the present disclosure;
(20) FIG. **18** is an exploded top perspective view of the wire connector of FIG. **17**, illustrating components of the low-profile wire connector including a supplemental wire connector, a cover, and a base;
(21) FIG. **19** is an enlarged top plan view of a portion of the cover and supplemental wire connector of FIG. **18** taken from detail **19** of FIG. **18**, and illustrating the conductors of the supplemental wire connector aligned to enter the cover;
(22) FIG. **20** is an enlarged top plan view of an aperture of the cover of FIG. **19** taken from detail **20** of FIG. **19**, and illustrating the aperture configured with a knockout;
(23) FIG. **21** is an enlarged top plan view of an aperture of the cover of FIG. **19** taken from detail **21** of FIG. **19**, and illustrating the aperture configured with a plug;
(24) FIG. **22** is a cross-sectional view of the wire connector of FIG. **17** taken from detail **22** of FIG. **17**, and illustrating a contact chamber of the cover holding an electrical contact which is electrically connecting three conductors;
(25) FIG. **23** is a bottom perspective view of the cover of the wire connector of FIG. **18**, illustrating electrical contacts removably held within the cover electrically connecting a conductor from the supplemental wire connector and two conductors of the wire connector;
(26) FIG. **24** is an enlarged view of an electrical contact positioned in the contact chamber of FIG. **23**, illustrating the contact chamber of the cover holding the electrical contact which is electrically connecting three electrical conductors;
(27) FIG. **25** is an enlarged top perspective view of an exemplary embodiment of an electrical contact of FIG. **23**, illustrating contact grips configured to electrically connect a conductor of the supplemental wire connector with the electrical contact;
(28) FIG. **26** is an enlarged top perspective view of an exemplary embodiment of an electrical contact of FIG. **23**, illustrating two contact fingers configured to electrically connect a conductor of the supplemental wire connector with the electrical contact;

(29) FIG. **27** is a side perspective view of an exemplary embodiment of a supplemental wire connector of FIG. **18**, illustrating the conductors of cable I inserted into a connector contact that is housed within the aperture;
(30) FIG. **27***a* is an enlarged cross-sectional view of the connector contact housed in the aperture of FIG. **27** taken from detail **27***a* of FIG. **27**, and illustrating the conductor being gripped by two teeth of the connector contact electrically connecting the conductor to the connector contact;
(31) FIG. **28** is a cross-sectional view of the wire connector of FIG. **17** taken from detail **28** of FIG. **17**, and illustrating a contact chamber of the cover holding an electrical contact configured to accommodate a connector contact so that the three conductors are electrically connected;
(32) FIG. **29** is a perspective view of an exemplary embodiment of a low-profile wire connector configured to accommodate multiple conductors per side of the wire connector;
(33) FIG. **30** is a perspective view of the wire connector of FIG. **29**, illustrating one side of the cover open and showing the conductors of cable Q and R sharing a corresponding electrical contact;
(34) FIG. **31** is a cross-section view of a portion of the wire connector of FIG. **29** taken from detail **31** of FIG. **29**, illustrating the electrical contact accommodating two conductors per side of the electrical contact;
(35) FIG. **32** is a perspective view of another exemplary embodiment of a low-profile connector assembly according to the present disclosure, illustrating two wire connectors of FIG. **17** coupled together; and
(36) FIG. **33** is an exploded perspective view of the low-profile connector assembly of FIG. **32**, illustrating cables and components of the two wire connectors including covers, bases, and supplemental wire connectors.

DETAILED DESCRIPTION

(37) The present disclosure provides descriptions of embodiments for low profile wire connectors used for splicing electrical conductors in cabling having at least one control conductor and at least one power conductor. This disclosure and the accompanying drawings are to be regarded in an illustrative sense rather than a restrictive sense. Various modifications may be made thereto without departing from the spirit and scope of the present disclosure.
(38) Referring to FIGS. **1-7**, an exemplary embodiment of a low-profile wire connector **10** according to the present disclosure is shown. The wire connectors **10** are configured to connect or splice electrical cables A and B, seen in FIG. **1**, to each other. The electrical cables A and B may include a set of electrical conductors rated for line voltages that supply electrical power, or the electrical cables A and B may include a set of electrical conductors rated for low voltages to provide control signals. As a non-limiting example, the line voltage conductors may be 10 AWG to 14 AWG conductors, and the low voltage conductors may be 18 AWG to 24 AWG conductors. In the exemplary embodiment shown, the cable A includes a set of electrical conductors A1, A2, A3 that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set. Similarly, cable B includes a set of electrical conductors B1, B2, B3 that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set.
(39) Continuing to refer to FIGS. **1-7**, in this exemplary embodiment, the wire connector **10** includes a base **20** and a cover **150**. The base **20** and cover **150** of each wire connector **10** are made of an electrical insulating material. As a non-limiting example, the insulating material for the base **20** and cover **150** may have an insulation factor that is the same insulation factor as the insulation jackets of the line voltage conductors. In addition to or another non-limiting example, is that the base **20** and cover **150** may be dimensioned to provide a minimum 0.25 of an inch physical separation between the low voltage conductors and the line voltage conductors when a first wire connector **10** splicing low voltage conductors is coupled to a second wire connector **10** splicing line voltage conductors. The wire connectors **10** may also include one or more cover coupling

assemblies **50**, seen in FIG. **2**A, and one or more base coupling assemblies, e.g., base coupling assemblies **320** and **330**, seen in FIGS. **11-14**. Each of the one or more cover coupling assemblies **50** is used to releasably or permanently secure a portion of the cover **150** to the base **20**. Each of the one or more base coupling assemblies, e.g., base coupling assemblies **320** and **330**, is used to releasably or permanently secure a base **20** of one wire connector **10** to a base **20** of another wire connector **10** as described in more detail hereinbelow.

(40) Referring to FIGS. **2-4**, the base **20** has a body **22** that in the embodiment shown is a substantially flat or planar structure or platform having an upper wall **22***a*, a lower wall **22***b*, and an outer wall **22***f*. For ease of description, the body **22** may be divided into a first end portion **22***c*, a second end portion **22***d* and a central portion **22***e* between the first end portion **22***c* and the second end portion **22***d*. The body **22** may be a solid structure or, as shown in FIGS. **3** and **4**, the body **22** may have a solid upper wall **22***a*, and a solid outer wall **22***f* with hollowed out portions that form one or more structural reinforcing ribs **22***g*, seen in FIG. **4**. In this example, a bottom surface of the outer wall **22***f* and the one or more structural reinforcing ribs **22***g* form the lower wall **22***b* of the body **22**. The hollowed-out portions reduce the material needed to manufacture the wire connectors **10** and thus reduce the cost to manufacture the wire connectors **10**. The thickness of the body **22** may vary for each wire connector **10** based upon a number of factors, including the type of wires that are being connected in the wire connector **10** and the NEC insulation specifications for separating low voltage conductors from line voltage conductors. As stated previously, a minimum 0.25 of an inch physical separation between the low voltage conductors and the high voltage conductors meets the NEC specification. The thickness of the body **22** of the wire connector **10** will be described in further detail hereinbelow.

(41) The base **20** may include one or more cable support members **24**, one or more pair of tabs **26**, one or more alignment apertures **28** and one or more mounting tabs **30**, seen in FIG. **3**. Each of the one or more cable support members **24** extends from the upper wall **22***a* of the body **22**. Preferably, one of the one or more cable support members **24** is positioned at or near an outer edge of the first end portion **22***c* of the body **22**, and another of the one or more cable support members **24** is positioned at or near an outer edge of the second end portion **22***d* of the body **22**, as shown in FIG. **3**. Each of the one or more cable support members **24** is configured to accommodate or fit at least a portion of an insulating jacket of a cable, e.g., cable A or cable B, and may include a curved surface or groove **24***a* that conforms to the shape of the cable A or cable B.

(42) Referring to FIG. **3**, the one or more pair of tabs **26** may assist in aligning the cover **150** to the base **20** when attaching the cover **150** to the base **20**. When the cover **150** is attached to the base **20**, each pair of tabs **26** may form a portion of a contact chamber **170**, as seen in FIG. **4**, in the cover **150** as described in more detail below. Each of the one or more pair of tabs **26** extend from the upper wall **22***a* of the central portion **22***e* of the body **22** and away from the body **22**. Preferably, each tab **26***a* and **26***b* in the pair of tabs **26** extend from the upper wall **22***a* of the body **22** so that each tab **26***a* and **26***b* is substantially perpendicular to the upper wall **22***a*. In the embodiment shown, each tab **26***a* and **26***b* in the pair of tabs **26** is substantially planar in shape. However, the shape of each tab **26***a* and **26***b* in the pair of tabs **26** may be in any shape sufficient to assist in aligning the cover **150** to the base **20** when attaching the cover **150** to the base **20**.

(43) Continuing to refer to FIG. **3**, each of the one or more alignment apertures **28** extends through or into the body **22**, preferably, in the central portion **22***e* of the body. The one or more alignment apertures **28** are described in more detail below. In the event the body **22** includes the one or more mounting tabs **30**, the mounting tabs **30** are integrally or monolithically formed into the outer wall **22***f* of the body **22** in the central portion **22***e* of the body **22**. It is noted however, that the mounting tabs **30** may be secured to the outer wall **22***f* of the body **22** using, for example, welds. The mounting tabs **30** include an aperture **32**, as seen in FIG. **4**, therethrough that facilitate the securing of the base **20** to the cover **150**. Preferably, an alignment aperture **28** is positioned adjacent to or in close proximity to a mounting tab **30** of the body **22**.

(44) Turning now to FIGS. **1** and **4-7**, in this exemplary embodiment the cover **150** includes a main cover member **152** and one or more movable cover members that are movably connected to the main cover member **152**. In the embodiment shown, the cover **150** includes two movable cover members **154** and **156** that are pivotably joined to the main cover member **152** using a living or integral hinge structure to form a unitary or monolithically structured cover **150**. As such, the movable cover members **154** and **156** are capable of moving between an open position, seen in FIG. **2**, and a closed position, seen in FIG. **1**. As shown in FIG. **1**, the movable cover member **154** is configured and dimensioned to align with the first end portion **22***c* of the body **20**, movable cover member **156** is configured and dimensioned to align with the second end portion **22***d* of the body **20**, and the main cover member **152** is configured and dimensioned to align with the central portion **22***e* of the body **20**. The main cover member **152** includes an upper wall **152***a*, a first side wall **152***b* having one end extending from one end of the upper wall **152***a* and a second side wall **152***c* having one end extending from another end of the upper wall **152***a* so that the upper wall **152***a* and the side walls **152***b* and **152***c* form a U-shaped structure with a hollow interior. The movable cover member **154** includes an upper wall **154***a*, a first side wall **154***b* having one end extending from one end of the upper wall **154***a*, a second side wall **154***c* having one end extending from another end of the upper wall **154***a* and an end wall **154***d* joined between the side walls **154***b* and **154***c* and joined to the upper wall **154***a*. In this configuration, the upper wall **154***a*, the side walls **154***b* and **154***c* and the end wall **154***d* form a structure with a hollow interior that can receive one or more low voltage conductors or one or more high voltage conductors. Similarly, the movable cover member **156** includes an upper wall **156***a*, a first side wall **156***b* having one end extending from one end of the upper wall **156***a* and a second side wall **156***c* having one end extending from another end of the upper wall **156***a* and an end wall **156***d* joined between the side walls **156***b* and **156***c* and joined to the upper wall **156***a*. In this configuration, the upper wall **156***a*, the side walls **156***b* and **156***c* and the end wall **156***d* form a structure with a hollow interior that can receive one or more low voltage conductors or one or more high voltage conductors. In this embodiment, the upper wall **154***a* of the movable cover member **154** is joined to the upper wall **152***a* of the main cover member **152** with a living hinge, and the upper wall **156***a* of the movable cover member **156** is joined to the upper wall **152***a* of the main cover member **152** with a living hinge so that the movable cover members **154** and **156** are pivotable between the open and closed positions relative to the main cover member **152**.

(45) The main cover member **152** may include one or mounting tabs **158** as shown in FIG. **5**. The mounting tabs **158** may be integrally or monolithically formed into the side walls **152***b* and/or **152***c* of the main cover member **152**. It is noted however, that the mounting tabs **158** may be secured to the side walls **152***b* and/or **152***c* of the main cover member **152** using, for example, welds. The mounting tabs **158** include an aperture **160** therethrough that facilitate the securing of the base **20** to the cover **150**. The main cover member **152** may also include one or more contact chambers **170**. The one or more contact chambers **170** are positioned within the hollow interior of the main cover member **152**. In the embodiment shown, there are three contact chambers **170**, where each contact chamber **170** is configured and dimensioned to house a single electrical contact **200**. More specifically, in the embodiment shown, each contact chamber **170** includes one or more walls **172** that define a cavity **174** in which the at least one electrical contact **200** is installed. The one or more walls **172** may be integrally or monolithically formed into main cover member **152** or the one or more walls **172** may be secured to the main cover member **152** using, for example, welds or mechanical fasteners. An example of a weld is an ultrasonic weld. Each contact chamber **170** may include one or more contact mounting structures **176** used for securing or releasably securing the at least one electrical contact **200** within the cavity **174** of the contact chamber **170**. In the exemplary embodiment shown in FIG. **5**, the contact mounting structures **176** are notches, channels or grooves in the walls **172** defining the particular contact chamber **170**. The one or more walls **172** of each contact chamber **170** may also include one or more openings **178** that act as access points into the

cavity **174** of the contact chamber **170**. Preferably, the one or more openings **178** are adjacent to or in close proximity to the electrical contact **200** within the cavity **174** of the contact chamber **170**. The openings **178** permit the low voltage or high voltage electrical conductors to pass through a contact chamber wall **172** to be electrically connected to the electrical contact **200** housed within the contact chamber **170**. Preferably, as shown in FIG. **5**, one opening **178** is disposed on one side of the contact chamber **170** so that electrical conductors in the first end portion **22***c* of the body **22** of the base **20** can be secured to an electrical contact **200**, and another opening **178** is disposed on the opposite side of the contact chamber **170** so that electrical conductors in the second end portion **22***d* of the body **22** of the base **20** can be secured to a corresponding electrical contact **200** within the contact chamber **170**. In addition, each opening **178** may be configured and dimensioned to receive one of the tabs **26***a* or **26***b* in the pair of tabs **26** extending from the body **22** of the base **20** to form a portion of the contact chamber wall **172**. Thus, when the cover **150** is coupled or secured to the base **20**, a tabs **26***a* or **26***b* is positioned between sides of the chamber walls **172** defining the opening **178** of the contact chamber **170** to at least partially block access to the electrical contact **200**. As such, each tabs **26***a* or **26***b* may be spaced from another tabs **26***a* or **26***b* a distance that is equal to a thickness of the chamber wall **172** between each of the openings **178** of the contact chamber **170**. Such a configuration allows the electrical contact **200** to be partly protected by one of the tabs **26***a* or **26***b* in the pair of tabs **26**. The embodiment exemplified by FIG. **5** illustrates the main cover member **152** having three contact chambers **170** positioned in series, and each contact chamber **170** includes two openings **178**. However, one skilled in the art would recognize that the main cover member **152** can have any number of contact chambers **170** which in turn may include any number of chamber wall openings **178**.

(46) Referring now to FIGS. **6** and **7**, an exemplary embodiment of an electrical contact **200** is shown. For ease of description, the electrical contact **200** may be divided into a first contact portion **202** and a second contact portion **204** that are interconnected such that an electrical path is formed between the first contact portion **202** and a second contact portion **204**. In the embodiment shown, the first contact portion **204** includes a portion of a contact plate **210**, a first stabilizing arm **212**, a first mounting arm **214** and a first contact arm **216**. Similarly, the second contact portion **204** includes a portion of a contact plate **210**, a second stabilizing arm **218**, a second mounting arm **220** and a second contact arm **222**. In the embodiment shown, the contact plate **210** is substantially planar plate where the first and second contact portions **202** and **204** are formed as a unitary or monolithic structure, and a portion of the contact plate **210** provides the electrical path between the first contact portion **202** and the second contact portion **204**. However, the present disclosure also contemplates that the first and second contact portions **202** and **204** may be formed as separate, spaced apart structures where the contact plates are electrically connected.

(47) The portion of the contact plate **210** within the first contact portion **202** includes a contact area **224**. The contact area **224** may include one or more contact enhancing surfaces **226** configured to increase the frictional force between the contact plate **210** and bare low voltage or high voltage electrical conductors positioned between the contact plate **210** and the first contact arm **216**. The contact enhancing surfaces **226** may be in the form of knurling, teeth or other surfaces capable of increasing the frictional force between the contact plate **210** and the electrical conductors. Non-limiting examples of other surfaces capable of increasing the frictional force include bumpy, coarse and/or gnarled surfaces. Similarly, the portion of the contact plate **210** within the second contact portion **204** includes a contact area **228**. The contact area **228** may include one or more contact enhancing surfaces **230** configured to increase the frictional force between the contact plate **210** and bare low voltage or high voltage electrical conductors, positioned between the contact plate **210** and the second contact arm **222**. The contact enhancing surfaces **230** may be in the form of knurling, teeth or other surfaces capable of increasing the frictional force between the contact plate **210** and the electrical conductors. Non-limiting examples of other surfaces capable of increasing the frictional force include bumpy, coarse and/or gnarled surfaces.

(48) Continuing to refer to FIGS. **6** and **7**, the first stabilizing arm **212** includes a first end **212***a* extending from a first side **210***a* of the contact plate **210** and is substantially perpendicular to the contact plate **210**. However, the first stabilizing arm **212** may be at an angle relative to the contact plate **210**. A first mounting arm **214** is at or near a second end **212***b* of the first stabilizing arm **212** and extends from a side of the first stabilizing arm **212** such that the first mounting arm **214** is substantially perpendicular to the first stabilizing arm **212**. However, first mounting arm **214** may be at an angle relative to the first stabilizing arm **212**. The first contact arm **216** extends from the first mounting arm **214** towards the contact plate **210**. The first contact arm **216** is configured such that the first contact arm **216** is able to flex, e.g., bend, when a conductor is inserted between the contact plate **210** and the first contact arm **216** permitting forward advancement of the conductor between the contact plate **210** and the first contact arm **216** while limiting or preventing withdrawal of the conductor from between the contact plate **210** and the first contact arm **216**. When a conductor is inserted between the contact plate **210** and the first contact arm **216**, the first contact arm **216** cooperates with the contact enhancing surface **226** to secure the conductor to the first contact portion **202** of the electrical contact **200**.
(49) Continuing to refer to FIGS. **6** and **7**, the second stabilizing arm **218** includes a first end **218***a* extending from a second side **210***b* of the contact plate **210** and is substantially perpendicular to the contact plate **210**. However, the second stabilizing arm **218** may be at an angle relative to the contact plate **210**. A second mounting arm **220** is at or near a second end **218***b* of the second stabilizing arm **218** and extends from a side of the second stabilizing arm **218** such that the second mounting arm **220** is substantially perpendicular to the second stabilizing arm **218**. However, second mounting arm **220** may be at an angle relative to the second stabilizing arm **218**. The second contact arm **222** extends from the second mounting arm **220** towards the contact plate **210**. The second contact arm **222** is configured such that the second contact arm **220** is able to flex, e.g., bend, when a conductor is inserted between the contact plate **210** and the second contact arm **222** permitting forward advancement of the conductor between the contact plate **210** and the second contact arm **222** while limiting or preventing withdrawal of the conductor from between the contact plate **210** and the second contact arm **222**. When a conductor is inserted between the contact plate **210** and the second contact arm **222**, the second contact arm **222** cooperates with the contact enhancing surface **230** to secure the conductor to the second contact portion **204** of the electrical contact **200**.
(50) As illustrated in FIGS. **5** and **7**, the electrical contacts **200** are fitted into the contact chamber **170**. Grooves in the walls **172** of the contact chamber **170** forming the contact mounting structure **176** are configured to receive the first mounting arm **214** and the second mounting arm **220**. Also shown in FIG. **7** is an example of two electrical conductors A1 and B1, e.g., two line voltage conductors or two low voltage conductors, connected to the electrical contact **170**. As shown in FIG. **7**, electrical conductor A1 is clamped between the first contact arm **216** and the contact enhancing surface **226** of the contact area **224** of the contact plate **210**. Friction from the contact enhancing surface **226** and the force applied by the first contact arm **216** against the electrical conductor A1 causes the electrical conductor A1 to be attached to the electrical contact **200**. Similarly, as shown in FIG. **7**, electrical conductor B1 is clamped between the second contact arm **222** and the contact enhancing surface **230** of the contact area **228** of the contact plate **210**. Friction from the contact enhancing surface **230** and the force applied by the second contact arm **222** against the electrical conductor B1 causes the electrical conductor B1 to be attached to the electrical contact **200**. As a result, an electrical connection between the conductors A1 and B1 is established.
(51) To permanently or releasably secure the cover **150** to the base **20**, the main cover member **152** is first aligned with the base **20**, seen in FIG. **4**, such that alignment pins **180** extending from one or more walls of the contact chambers **170**, seen in FIG. **5**, fit within the alignment apertures **28** in the body **22**, seen in FIGS. **3** and **4**. With the cover **150** properly aligned with the base **20**, the cover

150 is then permanently or releasably secured to the base **20**. To permanently secure the cover **150** to the base **20**, a mounting fastener (not shown), e.g., a rivet, can be passed through the aperture **32** in the mounting tab **30** of the body **22** and through the aperture **160** in the mounting tabs **158** on the main cover member **152**. The cover **150** may also be permanently secured to the base **20** using, for example, welds, such as ultrasonic welds. To releasably secure the cover **150** to base **20**, a mounting fastener (not shown), e.g., a nut and bolt, where the bolt can be passed through the aperture **32** in the mounting tab **30** of the body **22** and through the aperture **160** in the mounting tabs **158** on the main cover member **152** and the nut can be secured to the bolt.

(52) Referring again to FIGS. **2**, **3** and **5**, as described above, the two movable cover members **154** and **156** are pivotably joined to the main cover member **152** using a living or integral hinge structure. The movable cover member **154** is configured and dimensioned to align with the first end portion **22***c* of the body **20**, and the movable cover member **156** is configured and dimensioned to align with the second end portion **22***d* of the body **20**. To releasably secure the movable cover member **154** to the body **22** of the base **20** and the movable cover member **156** to the body **22** of the base **20**, one or more cover coupling assemblies **50** may be used. In the embodiment shown in FIG. **2**A, the one or more cover coupling assemblies **50** are snap fit joints. In this embodiment, two coupling assemblies **50** are used to releasably secure the movable cover member **154** to the body **22**, and two coupling assemblies **50** are used to releasably secure the movable cover member **156** to the body **22**. Each coupling member assembly **50** may include a cantilever beam **252** and a matching recess or opening **254**. The cantilever beam **252** extends from the body **22** of the base **20**, as shown in FIGS. **2** and **2**A, and has a free end **252***a*. The cantilever beam **252** includes a tapered hook **256** at its free end **252***a*. The matching recesses or openings **254** are located in the movable cover member **154** and the movable cover member **156**. The matching recesses **254** are positioned on the movable cover member **154** to align with the tapered hook **256** of the cantilever beam **252** when the movable cover member **154** is in a closed position, shown in FIG. **1**. Similarly, the matching recesses **254** are positioned on the movable cover member **156** to align with the tapered hook **256** of the cantilever beam **252** when the movable cover member **156** is in a closed position, shown in FIG. **1**.

(53) Referring now to FIGS. **8-12**, the two wire connectors **10** may be coupled together to form a wire connector assembly **300**. In the embodiment shown, each wire connector is substantially the same as the wire connectors **10** described above such that a detailed description of the wire connectors is not repeated. However, to distinguish between the two wire connectors in this exemplary embodiment, each wire connector is referenced with an alphanumeric identifier, such that a first of the two wire connectors is referenced as **10***x* and a second of the two wire connectors is referenced as **10***y*. In addition, certain components of the wire connectors **10***x* and **10***y* will follow the same alphanumeric identification, such that, for example, the base of wire connector **10***x* is referenced as base **20***x* and the cover of wire connector **10***x* is referenced as cover **150***x*, and the base of wire connector **10***y* is referenced as base **20***y* and the cover of wire connector **10***y* is referenced as cover **150***y*. Components of the bases **20***x* and **20***y* may have their original alphanumeric references, and components of the covers **150***x* and **150***y* may have their original alphanumeric references.

(54) In the exemplary embodiment of FIGS. **8-12**, the wire connector **10***x* receives and splices electrical conductors in line voltage cables C and D, and the wire connector **10***y* receives and splices electrical conductors in low voltage cables E and F. Each cable C and D may include a double insulated set of electrical conductors rated for line voltages, and each cable E and F may include a double insulated set of electrical conductors rated for low voltages. The line voltage cables C and D may be used to provide electrical power. The cable C includes a set of line voltage conductors C1, C2, C3 that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set. Similarly, the cable D includes a set of line voltage conductors D1, D2, D3 that are individually insulated by a first insulating sleeve or jacket

with a second insulating sleeve or jacket surrounding the set. The low voltage cables E and F may be used for control/signaling purposes. The cable E includes a set of low voltage conductors E1, E2, E3 that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set. Similarly, the cable F includes a set of low voltage conductors F1, F2, F3 that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set. It should also be noted that, while the embodiment shown presents the cables C and D as having line voltage conductors and the cables E and F as having low voltage conductors, one skilled in the art would recognize that any of the cables C, D, E, F can include either low voltage conductors or line voltage conductors.

(55) As shown in FIGS. **9-14**, to secure the wire connector **10***x* to the wire connector **10***y* one or more base coupling assemblies, e.g., **320** and/or **330** may be used. In the exemplary embodiment shown, the one or more base coupling assemblies **320** are snap fit joints. However, the present disclosure contemplates other fasteners to secure the wire connector **10***x* to the wire connector **10***y*. Non-limiting examples include nuts and bolts, and welds, such as ultrasonic welds. In the embodiment shown, base coupling assemblies **320** and **330** are snap fit joints that are used to secure wire connectors **10***x* and **10***y* together. The base coupling assembly **320** includes a cantilever beam **322** and a matching lip **324**. In the embodiment shown, the cantilever beam **322** extends from the upper wall **22***a* of the body **22** of the base **20***y* in a direction away from the cover **150***y*, as shown in FIGS. **10** and **11**. The cantilever beam **322** has a free end **322***a* that includes a tapered hook **326**. The matching lip **324** for the cantilever beam **322** is located in the lower wall **22***b* of the body **20***x* of the first wire connector **10***x*. The matching lip **324** is positioned on the lower wall **22***b* of the body **20***x* to align with the tapered hook **326** of the cantilever beam **322** when the base **20***x* of the first wire connector **10***x* is positioned to be coupled to the base **20***y* of the second wire connector **10***y*, as shown in FIGS. **11** and **12**. Base coupling assembly **330** includes a cantilever beam **332** and a matching lip **334**, seen in FIGS. **13** and **14**. In the embodiment shown, the cantilever beam **332** extends from the upper wall **22***a* of the body **22** of the base **20***x* in a direction away from the cover **150***x*, as shown in FIGS. **13** and **14**. The cantilever beam **332** has a free end **332***a* that includes a tapered hook **336** at its free end **332***a*. The matching lip **334** for the cantilever beam **332** is located in the lower wall **22***b* of the body **22** of the base **20***y*. The matching lip **334** is positioned on the lower wall **22***b* of the body **22** of the base **20***y* to align with the tapered hook **336** of the cantilever beam **332** when the base **20***x* of the first wire connector **10***x* is positioned to be coupled to the base **20***y* of the second wire connector **10***y*, as shown in FIGS. **13** and **14**.

(56) When coupling the first wire connector **10***x* to the second wire connector **10***y*, the base **20***x* of the first wire connector **10***x* is positioned adjacent the base **20***y* of the second wire connector **10***y* as shown in FIGS. **9** and **10** so that their respective cantilever beam **322** and **332** are aligned with their corresponding matching lips **324** and **334**. The base **20***x* of the first wire connector **10***x* is then pressed against the base **20***y* of the second wire connector **10***y* so that the base coupling assemblies **320** and **330** couple the bases **20***x* and **20***y* together.

(57) Referring now to FIGS. **15** and **16**, another exemplary embodiment of a low-profile wire connector assembly **350** according to the present disclosure is shown. In this exemplary embodiment, the wire connector assembly **350** is substantially the same as the wire connector assembly **300** described above, except that the first base **20***x* and the second base **20***y* are replaced by a single base **360**. The base **360** is substantially similar to the combination of bases **20***x* and **20***y* with the exception that there is a single body **362** and there are no base coupling assemblies **320** and **330**. In this embodiment, the thickness of the body **362** is preferably sufficient to provide a minimum 0.25 inch physical separation between the line voltage conductors in cables C and D and the low voltage conductors in cables E and F.

(58) Referring now to FIGS. **17-28**, an exemplary embodiment of a low-profile wire connector according to the present disclosure is shown. To help distinguish between this embodiment of the wire connector and the aforementioned embodiments of the wire connector, this disclosure will

continue the practice of referencing different embodiments hereinafter with an alphanumeric identifier. In addition, just as certain components followed the same alphanumeric identification, e.g., the base of wire connector **10***x* is referenced as base **20***x* and the cover of wire connector **10***x* is referenced as cover **150***x*, the ensuing reference to different embodiments disclosed will continue this practice.

(59) In this exemplary embodiment, portions of the wire connector **10***z*, e.g., cover **150***z* and base **20***z* are substantially similar to the cover **150** and base **20**, respectively, of the wire connector **10** described above with respect to FIGS. **1-7**. However, according to the present embodiment, the cover **150***z* and base **20***z* are configured for attaching a supplemental wire connector **70** that allows for an additional third cable, cable I, to connect to or splice to the two cables G and H. That is, as illustrated in FIG. **17**, the wire connector **10***z* connects or splices cables G, H, and I to each other. The electrical cables G, H, and I may include a set of electrical conductors rated for line voltages that supply electrical power, or the electrical cables G, H, and I may include a set of electrical conductors rated for low voltages to provide control signals. As a non-limiting example, the line voltage conductors may be 10 AWG to 14 AWG conductors, and the low voltage conductors may be 18 AWG to 24 AWG conductors. In the exemplary embodiment shown, the cable G includes a set of electrical conductors G1, G2, G3 that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set. Similarly, cable H includes a set of electrical conductors H1, H2, H3 that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set. Furthermore, cable I includes a set of electrical conductors I1, I2, I3 that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set.

(60) Continuing to refer to FIGS. **17-28**, the wire connector **10***z* includes a base **20***z*, a cover **150***z*, and may include a supplemental wire connector **70**. The base **20***z*, cover **150***z*, and supplemental wire connector **70** are made of an electrical insulating material. As a non-limiting example, the insulating material for the base **20***z*, cover **150***z*, and supplemental wire connector **70** may have an insulation factor that is the same insulation factor as the insulation jackets of the line voltage conductors. In addition to or another non-limiting example, is that the base **20***z*, cover **150***z*, and supplemental wire connector **70** may be dimensioned to provide a minimum 0.25 of an inch physical separation between the low voltage conductors and the line voltage conductors when a first wire connector splicing low voltage conductors is coupled to a second wire connector splicing line voltage conductors. The wire connectors **10***z* may also include one or more cover coupling assemblies **50**, similar to that described above with respect to FIG. **2**A. Although not shown, wire connectors **10***z* may include one or more base coupling assemblies, e.g., base coupling assemblies **320** and **330**, similar to that described above with respect to FIGS. **11-14**, and/or one or more base coupling assemblies, e.g., **520**, that will be described in more detail below. The supplemental wire connector **70** may include one or more supplemental wire connector coupling assemblies, e.g., **420**, shown in FIGS. **17** and **18** (only one side shown), and one or more supplemental wire connector cover coupling assemblies **90**, seen in FIG. **18**. Each of the one or more cover coupling assemblies **50** is used to releasably or permanently secure a portion of the cover **150***z* to the base **20***z* in a manner that is substantially the same as described above with respect to previous embodiments. Each of the one or more base coupling assemblies, e.g., base coupling assemblies **320** and **330** and/or base coupling assemblies **520**, may be used to releasably or permanently secure a base **20***z* of one wire connector **10***z* to a base **20***z* of another wire connector **10***z* in a manner that is substantially the same as described above with respect to previously described embodiments. Each of the one or more supplemental wire connector coupling assemblies, e.g., **420** is used to releasably or permanently secure a supplemental wire connector **70** to a cover **150***z* of a wire connector **10***z* as described in more detail below. Each of the one or more supplemental wire connector cover coupling assemblies **90** is used to releasably or permanently secure a portion of the cover **72** to the base **74** of supplemental wire connector **70** as described in more detail below.

(61) Referring to FIGS. **17-28**, the base **20**z is substantially similar to the above-described embodiments and for reasons of brevity will not be described in detail.

(62) Turning now specifically to FIGS. **18-21**, in this exemplary embodiment, the cover **150**z is substantially similar to cover **150** described with respect to previous embodiments, except that the main cover member **152** may have at least one aperture **153** for providing access to the electrical contact **300** therebelow for receiving a conductor which, depending on a particular embodiment, may be a wire, a blade-type connector, and/or pin type connector. extending from supplemental wire connector **70**. Similarly, to the previously described covers, the main cover **150**z includes a main cover member **152** and one or more movable cover members that are movably connected to the main cover member **152**. In the embodiment shown, the cover **150**z includes two movable cover members **154** and **156** that are pivotably joined to the main cover member **152** using a living or integral hinge structure to form a unitary or monolithically structured cover **150**z. As such, the movable cover members **154** and **156** are capable of moving between an open position, not shown, and a closed position as depicted in FIGS. **17** and **18**. As shown in FIG. **18**, the movable cover member **154** is configured and dimensioned to align with the first end portion **22***c* of the body **20**z, movable cover member **156** is configured and dimensioned to align with the second end portion **22***d* of the body **20**z, and the main cover member **152** is configured and dimensioned to align with the central portion **22***e* of the body **20**z.

(63) The main cover member **152** includes an upper wall **152***a*, a first side wall **152***b* having one end extending from one end of the upper wall **152***a* and a second side wall **152***c* having one end extending from another end of the upper wall **152***a* so that the upper wall **152***a* and the side walls **152***b* and **152***c* form a U-shaped structure with a hollow interior. The upper wall **152***a* may also have at least one aperture **153** for providing access to the electrical contact **300** therebelow (e.g., see FIG. **22**). The embodiment illustrated in FIGS. **18** and **19** has three apertures **153**. Depending on a particular embodiment, the apertures **153** could be wire receiving holes to accommodate conductor wires. For example, as shown in FIG. **22**, the stripped end I of conductor I3 extends directly from supplemental wire connector **70** through the main cover member **152** and is received in connector **300** in the unit below. According to other illustrative embodiments to be described below, blade receiving slots to accommodate a blade-type conductor or pin receiving holes to accommodate a pin-type conductor extending from supplemental wire connector **70** may be provided. Preferably, a way is provided for covering any unused apertures **153**. For example, knockouts **153***a*, seen in FIG. **20**, and/or plugs **153***b*, seen in FIG. **21**, may be used to cover the apertures **153** to prevent debris from entering the contact chamber **170** and disrupting electrical conduction. The knockouts **153***a* may be selectively “knocked out” if there is a need for an additional cable. Similarly, the plug **153***b* may be selectively removed if there is a need for an additional cable. The connection of a conductor with an electrical contact **300** through the at least one aperture **153** will be discussed in greater detail below.

(64) The movable cover member **154** includes an upper wall **154***a*, a first side wall **154***b* having one end extending from one end of the upper wall **154***a*, a second side wall **154***c* having one end extending from another end of the upper wall **154***a* and an end wall **154***d* joined between the side walls **154***b* and **154***c* and joined to the upper wall **154***a*. In this configuration, the upper wall **154***a*, the side walls **154***b* and **154***c* and the end wall **154***d* form a structure with a hollow interior that can receive one or more low voltage conductors or one or more high voltage conductors. Similarly, the movable cover member **156** includes an upper wall **156***a*, a first side wall **156***b* having one end extending from one end of the upper wall **156***a* and a second side wall **156***c* having one end extending from another end of the upper wall **156***a* and an end wall **156***d* joined between the side walls **156***b* and **156***c* and joined to the upper wall **156***a*. In this configuration, the upper wall **156***a*, the side walls **156***b* and **156***c* and the end wall **156***d* form a structure with a hollow interior that can receive one or more low voltage conductors or one or more high voltage conductors. In this embodiment, the upper wall **154***a* of the movable cover member **154** is joined to the upper wall

152*a* of the main cover member **152** with a living hinge, and the upper wall **156***a* of the movable cover member **156** is joined to the upper wall **152***a* of the main cover member **152** with a living hinge so that the movable cover members **154** and **156** are pivotable between the open and closed positions relative to the main cover member **152**.
(65) The main cover member **152** may also include one or more contact chambers **170** (e.g., see FIG. **23**). The one or more contact chambers **170** are positioned within the hollow interior of the main cover member **152**. In the embodiment shown, there are three contact chambers **170**, where each contact chamber **170** is configured and dimensioned to house a single electrical contact **300**. More specifically, in the embodiment shown, each contact chamber **170** includes one or more walls **172** that define a cavity **174** in which the at least one electrical contact **300** is installed. The one or more walls **172** may be integrally or monolithically formed into main cover member **152** or the one or more walls **172** may be secured to the main cover member **152** using, for example, welds or mechanical fasteners. An example of a weld is an ultrasonic weld. Each contact chamber **170** may include one or more contact mounting structures **176** used for securing or releasably securing the at least one electrical contact **300** within the cavity **174** of the contact chamber **170**. In the exemplary embodiment shown in FIGS. **23** and **24**, the contact mounting structures **176** are notches, channels or grooves in the walls **172** defining the particular contact chamber **170**. The one or more walls **172** of each contact chamber **170** may also include one or more openings **178** that act as access points into the cavity **174** of the contact chamber **170**. Preferably, the one or more openings **178** are adjacent to or in close proximity to the electrical contact **300** within the cavity **174** of the contact chamber **170**. The openings **178** permit the low voltage or high voltage electrical conductors to pass through a contact chamber wall **172** to be electrically connected to the electrical contact **300** housed within the contact chamber **170**. Preferably, as shown in FIG. **23**, one opening **178** is disposed on one side of the contact chamber **170** so that electrical conductors in the first end portion **22***c* of the body **22** of the base **20***z* can be secured to an electrical contact **300**, and another opening **178** is disposed on the opposite side of the contact chamber **170** so that electrical conductors in the second end portion **22***d* of the body **22** of the base **20***z* can be secured to a corresponding electrical contact **300** within the contact chamber **170**. In addition, each opening **178** may be configured and dimensioned to receive one of the tabs **26***a* or **26***b* in the pair of tabs **26** extending from the body **22** of the base **20**z (e.g., see FIG. **18**) to form a portion of the contact chamber wall **172**. Thus, when the cover **150**z is coupled or secured to the base **20***z*, the tabs **26***a* and **26***b* is positioned between sides of the chamber walls **172** defining the opening **178** of the contact chamber **170** to at least partially block access to the electrical contact **200**. As such, each tab **26***a* and **26***b* may be spaced from another tab **26***a* or **26***b* a distance that is equal to a thickness of the chamber wall **172** between each of the openings **178** of the contact chamber **170**. Such a configuration allows the electrical contact **300** to be partly protected by one tabs **26***a* and **26***b* in the pair of tabs **26**. The embodiment exemplified by FIG. **23** illustrates the main cover member **152** having three contact chambers **170** positioned in series, and each contact chamber **170** includes two openings **178**. However, one skilled in the art would recognize that the main cover member **152** can have any number of contact chambers **170** which in turn may include any number of chamber wall openings **178**.
(66) Referring again to FIGS. **17** and **18**, the supplemental wire connector **70** includes a cover member **72** and a body **74**. The cover member **72** includes an upper wall **72***a*, a first side wall **72***b* having one end extending from one end of the upper wall **72***a*, a second side wall **72***c* having one end extending from another end of the upper wall **72***a*, and an end wall **72***d* joined between the side walls **72***b* and **72***c* and joined to the upper wall **72***a*. In this configuration, the upper wall **72***a*, the side walls **72***b* and **72***c*, and the end wall **72***d* form a structure with a hollow interior that can receive one or more low voltage conductors or one or more high voltage conductors. The body **74** is substantially planar and has a bottom wall **76** that extends away from the body **74** at one end. The bottom wall **76** may be integrally or monolithically formed into one end of the body **74**. It is

also noted, however, that the bottom wall **76** may be secured to one end of the body **74** using, for example, welds. The cover member **72** is movably connected to the bottom wall **76** using a living or integral hinge structure to form a unitary or monolithically structured supplemental wire connector **70**. The bottom wall **76** has at least one wire receiving aperture **78**. The embodiment illustrated in FIG. **18** has three wire receiving apertures **78** in series that align with the apertures **153** of the upper wall **152***a* of the main cover member **152** of the cover **150***z*, as shown in FIG. **19**, that allow a conductor to communicate with an electrical contact **300** that resides in the contact chamber **170**.

(67) As noted above, the cover member **72** of the supplemental wire connector **70** is pivotably joined to the bottom wall **76** using a living or integral hinge structure. The movable cover member **72** of the supplemental wire connector **70** is dimensioned to align with the body **74** of the supplemental wire connector **70**. To releasably secure the movable cover member **72** to the body **74**, one or more supplemental cover coupling assemblies **90** may be used. In the embodiment shown in FIG. **18**, the one or more supplemental cover coupling assemblies **90** are snap fit joints. In this embodiment, two supplemental cover coupling assemblies **90** are used to releasably secure the movable cover member **72** of the supplemental wire connector **70** to the body **74** of the supplemental wire connector **70**. Each supplemental cover coupling assembly **90** may include a cantilever beam **482** and a matching recess or opening **484**. The cantilever beam **482** extends from the body **74** of the supplemental wire connector **70**, as shown in FIG. **18**, and has a free end **482***a*. The cantilever beam **482** includes a tapered hook **486** at its free end **482***a*. The matching recesses or openings **484** are located in the side walls, **72***b* and **72***c*, of the cover member **72** and positioned to align with the tapered hook **486** of the cantilever beam **482** when the cover member **72** of the supplemental wire connector **70** is in a closed position, shown in FIG. **17**.

(68) A supplemental wire connector **70** according to an illustrative embodiment of the present disclosure is described in more detail by reference to FIGS. **27**, **27**A. In this embodiment, the wire receiving apertures **78** formed in the bottom wall **76** of supplemental wire connector **70** include a connector contact **80** that resides within each wire receiving aperture **78**. The connector contact **80** may be secured in the wire receiving aperture **78** by friction fit, adhesives, or mechanical fasteners. For example, as shown, one or more tines **83** may be provided around the periphery of contact **80** for helping secure contact **80** in receiving aperture **78**. The connector contact **80** includes a hollow cylindrical body **82**, a collar **84**, and a contact **86**. The connector contact **80** is made of a material that is electrically conductive. Non-limiting examples of electrically conductive material that may be used include but are not limited to, for example, copper, copper alloy, phosphor bronze, aluminum or brass. For ease of description, the body **82** of the connector contact **80** may be divided into a first end **82***a* and a second end **82***b*. The first end **82***a* of the body **82** is configured to accommodate a conductor, e.g., a wire, to be inserted therein. For example, where supplemental wire connector **70** is to accommodate line voltage conductors, the first end **82***a* may be dimensioned to receive line voltage conductors generally in the 10 AWG to 14 AWG range. Where supplemental wire connector **70** is to accommodate low voltage conductors, the first end **82***a* may be dimensioned to receive low voltage conductors generally in the 18 AWG to 24 AWG range. The collar **84** may be integrally or monolithically formed into the body **82** of the connector contact **80** at generally the first end **82***a*. The collar **84** may also be secured to the connector contact **80** by other means, for example, welds. The contact **86** may be integrally or monolithically formed into the second end **82***b* of the body **82** of the connector contact **80** so that the contact **86** extends away from the first end **82***a* of the body **82** a sufficient distance to allow the contact **86** to be able to pass through the apertures **153** of the main cover member **152** and interact with the electrical contact **300** residing in the contact chamber **170**. It is contemplated that the contact **86** may be attached to the second end **82***b* of the body **82** by other means including, for example, welds. The contact **86** may be a pin-type or a blade-type contact. The interior of the body **82** may also include at least one tooth **88** configured to allow the advancement of a conductor into the connector contact **80** while

limiting or preventing the withdrawal of the conductor from the connector contact **80**. The at least one tooth **88** may also pierce the outer diameter of the inserted conductor to ensure electrical conductivity is maintained between the connector contact **80** and the inserted conductor. According to the non-limiting embodiment illustrated in FIG. **27**A, the interior of the body **82** has two teeth **88** for gripping the inserted conductor and maintaining electrical conductivity between the conductor and the connector contact **80**.

(69) As depicted in FIG. **28**, the spliced end of conductor I3 is press fit into connector contact **80** of supplemental wire connector **70** and contact **86** extends from the bottom thereof. When supplemental wire connector **70** is attached to cover **152**, contact **86** extends through the orifice in cover **152** and connects to electrical contact **300** as shown. Alternatively, or in addition, it will be appreciated that contacts **80** may be provided in cover **152** so that contact **86** extends into electrical contact **300** and such that the spliced end of conductor I3 or the contact **86** extending from supplemental wire connector **70** (e.g., see FIG. **22**) is received in connector contact **80**.

(70) Referring now to FIGS. **24-26**, an exemplary embodiment of an electrical contact **300** and **400** are shown. The electrical contacts **300** and **400** illustrated in the FIGS. **24-26** are substantially similar to the electrical contact **200** described above with respect to FIGS. **6-7**. Similar element numbers for similar portions of electrical contacts **300**, **400** and **200** are used. However, according to the present illustrative embodiment depicted in FIG. **24**, the electrical contact **300** has an opening **306** in the contact plate **310**. According to the present illustrative embodiment depicted in FIG. **26**, the electrical contact **400** has an opening **406** in the contact plate **410**.

(71) For ease of description, the electrical contact **300** may be divided into a first contact portion **202** and a second contact portion **204** that are interconnected such that an electrical path is formed between the first contact portion **202** and a second contact portion **204**. In the embodiment shown, the first contact portion **202** includes a portion of a contact plate **310**, a first stabilizing arm **212**, a first mounting arm **214** and a first contact arm **216**. Similarly, the second contact portion **204** includes a portion of a contact plate **310**, a second stabilizing arm **218**, a second mounting arm **220** and a second contact arm **222**. In the embodiment shown, the first and second contact portions **202** and **204** are formed as a unitary or monolithic structure, and a portion of the contact plate **310** provides the electrical path between the first contact portion **202** and the second contact portion **204**.

(72) The portion of the contact plate **310** within the first contact portion **202** has a contact area including one or more contact enhancing surfaces **226** configured to increase the frictional force between the contact plate **310** and bare low voltage or high voltage electrical conductors positioned between the contact plate **310** and the first contact arm **216**. The contact enhancing surfaces **226** may be in the form of knurling, teeth or other surfaces capable of increasing the frictional force between the contact plate **310** and the electrical conductors. Non-limiting examples of other surfaces capable of increasing the frictional force include bumpy, coarse and/or gnarled surfaces. Similarly, the portion of the contact plate **310** within the second contact portion **204** has a contact area including one or more contact enhancing surfaces **230** configured to increase the frictional force between the contact plate **310** and bare low voltage or high voltage electrical conductors, positioned between the contact plate **310** and the second contact arm **222**. The contact enhancing surfaces **230** may be in the form of knurling, teeth or other surfaces capable of increasing the frictional force between the contact plate **310** and the electrical conductors. Non-limiting examples of other surfaces capable of increasing the frictional force include bumpy, coarse and/or gnarled surfaces.

(73) The first stabilizing arm **212** includes a first end **212***a* extending from a first side **210***a* of the contact plate **310** and is substantially perpendicular to the contact plate **210**. However, the first stabilizing arm **212** may be at an angle relative to the contact plate **310**. A first mounting arm **214** is at or near a second end **212***b* of the first stabilizing arm **212** and extends from a side of the first stabilizing arm **212** such that the first mounting arm **214** is substantially perpendicular to the first

stabilizing arm **212**. However, first mounting arm **214** may be at an angle relative to the first stabilizing arm **212**. The first contact arm **216** extends from the first mounting arm **214** towards the contact plate **310**. The first contact arm **216** is configured such that the first contact arm **216** is able to flex, e.g., bend, when a conductor is inserted between the contact plate **310** and the first contact arm **216** permitting forward advancement of the conductor between the contact plate **310** and the first contact arm **216** while limiting or preventing withdrawal of the conductor from between the contact plate **310** and the first contact arm **216**. When a conductor is inserted between the contact plate **310** and the first contact arm **216**, the first contact arm **216** cooperates with the contact enhancing surface **226** to secure the conductor to the first contact portion **202** of the electrical contact **300**.

(74) The second stabilizing arm **218** includes a first end **218***a* extending from a second side **210***b* of the contact plate **310** and is substantially perpendicular to the contact plate **310**. However, the second stabilizing arm **218** may be at an angle relative to the contact plate **310**. A second mounting arm **220** is at or near a second end **218***b* of the second stabilizing arm **218** and extends from a side of the second stabilizing arm **218** such that the second mounting arm **220** is substantially perpendicular to the second stabilizing arm **218**. However, second mounting arm **220** may be at an angle relative to the second stabilizing arm **218**. The second contact arm **222** extends from the second mounting arm **220** towards the contact plate **310**. The second contact arm **222** is configured such that the second contact arm **220** is able to flex, e.g., bend, when a conductor is inserted between the contact plate **310** and the second contact arm **222** permitting forward advancement of the conductor between the contact plate **310** and the second contact arm **222** while limiting or preventing withdrawal of the conductor from between the contact plate **310** and the second contact arm **222**. When a conductor is inserted between the contact plate **310** and the second contact arm **222**, the second contact arm **222** cooperates with the contact enhancing surface **230** to secure the conductor to the second contact portion **204** of the electrical contact **200**.

(75) The opening **306** is a generally square or rectangular opening, however, any shape of the opening that allows for a conductor to pass through the opening **306** and into the electrical contact **300** may be suitable. The opening **306** is preferably centrally located on the contact plate **310**. In the embodiment shown in FIG. **25**, the opening **306** has a first side **306***a*, a second side **306***b*, a third side **306***c*, and a fourth side **306***d*. The opening **306** may include contact grips, e.g., **307***a* and **307***b*, configured to allow the advancement of a conductor therethrough while limiting or preventing its withdrawal from the electrical contact **200**. In particular, the first contact grip **307***a* extends from the first side **306***a* and a second contact grip **307***b* extends from the third side **306***c*. The contact grips, **307***a* and **307***b*, are configured such that the contact grips are able to flex, e.g., bend, when a conductor is inserted between the two contact grips, **307***a* and **307***b*, permitting advancement of the conductor between the two contact grips while preventing withdrawal of the conductor from between the first contact grip **307***a* and the second contact grip **307***b*.

(76) An electrical contact according to another illustrative embodiment of the present disclosure is shown in FIG. **26** and is referred to as contact **400**. Contact **400** includes an opening **406** which is generally centrally located on the contact plate **410**. that allows for a conductor to pass through the opening **306** and into the electrical contact **300** may be suitable The opening **406** is a generally square or rectangular opening, however, any shape of the opening that allows for a conductor to pass through opening **406** and into electrical contact **400** may be suitable. In the present embodiment, the opening **406** has a first side **406***a*, a second side **406***b*, a third side **406***c*, and a fourth side **406***d*. The opening **406** may include contact grips, e.g., **408***a* and **408***b*, configured to allow the advancement of a conductor therethrough while limiting or preventing its withdrawal from the electrical contact **400**. In particular, the first contact grip **408***a* extends from the second side **406***b* and a second contact grip **408***b* extends from the fourth side **406***d*. The contact grips, **408***a* and **408***b*, are configured such that the contact grips are able to flex, e.g., bend, when a conductor is inserted between the two contact grips, **408***a* and **408***b*, permitting advancement of the

conductor between the two contact grips while preventing withdrawal of the conductor from between the first contact grip **408***a* and the second contact grip **408***b*.

(77) As illustrated in FIGS. **22-24**, the electrical contact **300** (or **400**) may be fitted into the contact chamber **170**. Grooves in the walls **172** of the contact chamber **170** forming the contact mounting structure **176** are configured to receive the first mounting arm **214** and the second mounting arm **220**. Also shown in FIG. **24** is an example of three electrical conductors G3, H3, and I3, (e.g., three-line voltage conductors or three low voltage conductors) connected to the electrical contact **300**. Electrical conductor H3 is clamped between the first contact arm **216** and the contact enhancing surface **226** of the contact area of the contact plate **310**. Friction from the contact enhancing surface **226** and the force applied by the first contact arm **216** against the electrical conductor H3 causes the electrical conductor H3 to be attached to the electrical contact **300**. Electrical conductor G3 is clamped between the second contact arm **222** and the contact enhancing surface **230** of the contact area of the contact plate **310**. Friction from the contact enhancing surface **230** and the force applied by the second contact arm **222** against the electrical conductor G3 causes the electrical conductor G3 to be attached to the electrical contact **300**. Electrical conductor I3 is clamped between the first contact grip **307***a* and the second contact grip **307***b*. The force applied by both contact grips, **307***a* and **307***b*, against the electrical conductor I3 causes the electrical conductor I3 to be electrically connected to the electrical contact **300**. As a result, an electrical connection between the conductors G3, H3, and I3 is established.

(78) Now referring again to FIGS. **17** and **18**, to secure the supplemental wire connector **70** with the cover **150***z* of the wire connector **10***z* one or more supplemental wire connector coupling assemblies, e.g., **420**, may be used. In the exemplary embodiment shown, the one or more supplemental wire connector coupling assemblies **420**, are snap fit joints. However, the present disclosure contemplates other fasteners to secure the supplemental wire connector **70** to the cover **150***z* of the wire connector **10***z*. Non-limiting examples include nuts and bolts, and welds, such as ultrasonic welds. In the embodiment shown, supplemental wire connector coupling assemblies **420** are snap fit joints that are used to secure the supplemental wire connector **70** to the cover **150***z* of the wire connector **10***z*. The supplemental wire connector coupling assembly **420** includes a cantilever beam **422** and a matching lip **424**. As shown in FIG. **18**, the cantilever beam **422** extends from the bottom wall **76** of the supplemental wire connector **70** in a direction away from the supplemental wire connector **70**. The cantilever beam **422** has a free end that includes a tapered hook **426**. A matching lip **424** which receives tapered hook **426** is located on the first side wall **152***c* of the main cover member **152** of the cover **150***z*. The matching lip **424** is positioned on the first side wall **152***c* of the main cover member **152** of the cover **150***z* to align with the tapered hook **426** of the cantilever beam **422** when the supplemental wire connector **70** is positioned to be coupled to the cover **150***z* of the wire connector **10***z*. Although not shown in FIGS. **17** and **18**, a similar coupling assembly **420** may be provided on the opposite sides of supplemental wire connector **70** and cover **150***z*.

(79) When coupling the supplemental wire connector **70** to the cover **150***z* of the wire connector **10***z*, the supplemental wire connector **70** is positioned adjacent the cover **150***z* of the wire connector **10***z* as shown in FIG. **18** so the respective cantilever beams **422** are aligned with their corresponding matching lips **424**. The supplemental wire connector **70** is then pressed against the cover **150***z* of the wire connector **10***z* so that the conductors extending from supplemental wire connector **70** are seated within connector **300** (or **400**) and so that the supplemental wire connector coupling assemblies **420** couple the supplemental wire connector **70** and cover **150***z* locking them together.

(80) Referring to FIGS. **29-31**, an exemplary embodiment of a wire connector **10***f* is shown. The wire connector **10***f* is configured to connect or splice cables Q and R to cables S and T. The electrical cables Q, R, S, and T may include a set of electrical conductors rated for line voltages that supply electrical power, or the electrical cables Q, R, S, and T may include a set of electrical

conductors rated for low voltages to provide control signals. As a non-limiting example, the line voltage conductors may be 10 AWG to 14 AWG conductors, and the low voltage conductors may be 18 AWG to 24 AWG conductors. In the exemplary embodiment shown, the cable Q includes a set of electrical conductors Q1, Q2, Q3 that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set. Similarly, the cable R includes a set of electrical conductors R1, R2, R3 that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set. Furthermore, the cable S includes a set of electrical conductors S1, S2, S3 that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set. Likewise, the cable T includes a set of electrical conductors T1, T2, T3 that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set.
(81) Continuing to refer to FIGS. **29-31**, in this exemplary embodiment, the wire connector **10***f* includes a base **20***f* and a cover **150***f* The base **20***f* is substantially the same as the base **10** described above with respect to previous embodiments and for the sake of brevity will not be described again here. The cover **150***f* is substantially similar to the cover **150** described above with respect to previous embodiments, except that the first side wall **154***b*, the second side wall **154***c*, and the end wall **154***d* of the movable cover member **154** of the cover **150***f* are taller to accommodate the additional cable (e.g., cable Q). In addition, depending on a particular embodiment, the first side wall **156***b*, the second side wall **156***c*, and the end wall **156***d* of the movable cover member **156** of the cover **150***f* may also be taller to accommodate the additional cable (e.g., cable S). As seen in FIG. **29**, the taller side walls, **154***b*, **154***c*, **156***b*, and **156***c*, as well as the taller end walls **154***d* and **156***d* create a vertical oblong opening that is capable of accommodating more than one cable in the first end portion **22***c* of the body **20** of wire connector **10***f* and similarly, in the second end portion **22***d* of the body of wire connector **10***f* FIG. **31** shows a partial cutaway view of FIG. **29** showing how two wires, R3 and Q3, from two cables, R and Q, respectively, may be inserted between the contact plate **210** and the first contact arm **216** of an electrical contact **200** and how two wires, S3 and T3 from two cables S and T, respectively, may be inserted between the contact plate **210** and the second contact arm **222** (not shown) so that all four conductors R3, Q3, S3 and T3 are electrically connected.
(82) Referring now to FIG. **32**, just as the previous embodiment wire connectors **10** described above with respect to FIGS. **1-7** may be coupled together, two wire connectors **10***r* and **10***t* may also be coupled together along with supplemental cable connectors **70***a* and **70***b* as shown, to form a wire connector assembly **500**. In the embodiment shown, each wire connector **10***r* and **10***t* is substantially the same as the wire connectors **10***z* described above with respect to FIGS. **17-28** such that a detailed description of the wire connectors is not repeated.
(83) In the exemplary embodiment of FIGS. **32-33**, the wire connector **10***r* and supplemental connector **70***a* receive and splice electrical conductors in line voltage cables J, K, and L. The wire connector **10***t* and supplemental connector **70***b* receive and splice electrical conductors in low voltage cables M, N, and O. Each cable J, K, and L may include a double insulated set of electrical conductors rated for line voltages, and each cable M, N, and O may include a double insulated set of electrical conductors rated for low voltages. The line voltage cables J, K, and L may be used to provide electrical power. The cable J includes a set of line voltage conductors (not shown) that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set. Similarly, the cable K includes a set of line voltage conductors (not shown) that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set. Furthermore, the cable L includes a set of line voltage conductors (not shown) that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set. The low voltage cables M, N, and O may be used for control/signaling purposes. The cable M includes a set of low voltage conductors (not shown) that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve

or jacket surrounding the set. Similarly, the cable N includes a set of low voltage conductors (not shown) that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set. Furthermore, the cable O includes a set of low voltage conductors (not shown) that are individually insulated by a first insulating sleeve or jacket with a second insulating sleeve or jacket surrounding the set.

(84) As shown in FIGS. **32** and **33**, to secure the wire connector **10***r* to the wire connector **10***t*, one or more base coupling assemblies **520** may be used. In the exemplary embodiment shown, the one or more base coupling assemblies **520** are snap fit joints. However, the present disclosure contemplates other fasteners to secure the wire connector **10***r* to the wire connector **10***t*. Non-limiting examples include nuts and bolts, and welds, such as ultrasonic welds. It is further contemplated that base coupling assemblies **320** and **330** described above with respect to FIGS. **11-14** may be used in place of base coupling assemblies **520** or in conjunction with base coupling assemblies **520**. In the embodiment shown, base coupling assemblies **520** are snap fit joints that are used to secure wire connectors **10***r* and **10***t* together. The base coupling assemblies **520** may include snap fit joints which include cantilever beam **522** including hook **526** and corresponding lip **524**. In the embodiment shown, the cantilever beam **522** extends from the outer wall **22***f* of the body **22** of the base **20***r* in a direction away from the cover **150***r*, as shown in FIG. **33**. The cantilever beam **522** may be integrally or monolithically formed into the outer wall **22***f* of the body **22** of the base **20***r*. It is noted however, that the cantilever beam **522** may be secured to the outer wall **22***f* of the body **22** using, for example, welds. The cantilever beam **522** has a free end that includes a tapered hook **526**. The corresponding lip **524** for receiving the tapered hook **526** is located in the outer wall **22***h* of the body **20***t* of the first wire connector **10***t*. The lip **524** is positioned on the outer wall **22***h* of the body **20***t* to align with the tapered hook **526** of the cantilever beam **522** when the base **20***t* of the first wire connector **10***t* is positioned to be coupled to the base **20***r* of the second wire connector **10***r*, as shown in FIGS. **32** and **33**. Although not shown in FIGS. **32** and **33**, a similar base coupling assembly **520** may be provided on the opposite side of base **20***r* and base **20***t*.

(85) When coupling the first wire connector **10***t* to the second wire connector **10***r*, the base **20***t* of the first wire connector **10***t* is positioned adjacent the base **20***r* of the second wire connector **10***r* as shown in FIGS. **32** and **33** so that the respective cantilever beams, **522**, are aligned with their corresponding matching lips **524**. The base **20***t* of the first wire connector **10***t* is then pressed against the base **20***r* of the second wire connector **10***r* so that the base coupling assemblies **320** couple the bases **20***t* and **20***r* together. The supplemental wire connectors **70***a*, **70***b* are attached to the first and second wire connectors **10***t*, **10***r* in a manner similar to that described above with respect to other embodiments.

(86) While illustrative embodiments of the present disclosure have been described and illustrated above, it should be understood that these are exemplary of the disclosure and are not to be considered as limiting. Additions, deletions, substitutions, and other modifications can be made without departing from the spirit or scope of the present disclosure. Accordingly, the present disclosure is not to be considered as limited by the foregoing description. For example, because the first wire connector **10***x* and the second wire connector **10***y* may splice different size electrical conductors, components of the first wire connector **10***x* may differ in size from components of the second wire connector **10***y*. To accommodate the larger line voltage conductors, each single electrical contact **200** and associated contact chamber **170** may be larger than their counterparts for low voltage conductors.

Claims

1. A wire connector assembly comprising: a first wire connector including: a first base having a first end portion, a second end portion and a central portion, the first base having an upper wall and a lower wall; a first cover having a main cover member, a first movable cover member movably

connected to a first side of the main cover member and a second movable cover member movably connected to a second side of the main cover member, the first cover being attached to the first base such that the main cover member of the first cover is associated with the central portion of the first base, the first movable cover member of the first cover is associated with the first end portion of the first base and the second movable cover member of the first cover is associated with the second end portion of the first base, the main cover member of the first cover including a plurality of contact chambers, each contact chamber of the main cover member of the first cover including at least one electrical contact positioned within a cavity of the contact chamber, the electrical contact providing at least two entry for conductors positioning from opposite direction, a first passage to the first end portion of the first base and a second passage to the second end portion of the first base; a first cover coupling assembly associated with the first end portion of the first base and the first movable cover member of the first cover and used to releasably couple the first movable cover member of the first cover to the first end portion of the first base; and a second cover coupling assembly associated with the second end portion of the first base and the second movable cover member of the first cover and used to releasably couple the second movable cover member of the first cover to the first end portion of the first base; and a second wire connector including: a second base having a first end portion, a second end portion and a central portion, the second base having an upper wall and a lower wall; a second cover having a main cover member, a first movable cover member movably connected to a first side of the main cover member and a second movable cover member movably connected to a second side of the main cover member, the second cover being attached to the second base such that the main cover member of the second cover is associated with the central portion of the second base, the first movable cover member of the second cover is associated with the first end portion of the second base and the second movable cover member of the second cover is associated with the second end portion of the second base, the main cover member of the second cover including a plurality of contact chambers, each contact chamber of the main cover member of the second cover including at least one electrical contact positioned within a cavity of the contact chamber, the electrical contact providing at least two entry for conductors positioning from opposite direction, a first passage to the first end portion of the second base and a second passage to the second end portion of the second base; a third cover coupling assembly associated with the first end portion of the second base and the first movable cover member of the second cover and used to releasably couple the first movable cover member of the second cover to the first end portion of the second base; and a fourth cover coupling assembly associated with the second end portion of the second base and the second movable cover member of the second cover and used to releasably couple the second movable cover member of the second cover to the first end portion of the second base; wherein the first base of the first wire connector is attached to the second base of the second wire connector.

2. The wire connector assembly of claim 1, wherein the main cover member further comprises at least one aperture configured to provide access to the electrical contact within the contact chamber.

3. The wire connector assembly of claim 2, wherein the at least one aperture further comprises a knockout configured to be removed to allow access therethrough.

4. The wire connector assembly of claim 2, wherein the at least one aperture further comprises a plug configured to be removed to allow access therethrough.

5. The wire connector assembly of **2**, further comprising a supplemental wire connector having a body and a movable cover movably connected to the body, the body including at least one aperture configured to receive a conductor; a supplemental wire connector coupling assembly associated with the body and the movable cover member of the supplemental wire connector and used to releasably couple the movable cover member of the supplemental wire connector to the body; and wherein the supplemental wire connector is configured to couple with the main cover member so that the at least one aperture of the main cover member align with the at least one aperture of the supplemental wire connector.

6. The wire connector of claim 5, wherein the at least one aperture houses a connector contact configured to receive and retain a conductor so that the conductor is electrically connected with the connector contact.
7. The wire connector of claim 6, wherein the connector contact is at least one of a pin-type and a blade-type connector contact.
8. The wire connector of claim 1, wherein the first movable cover is configured to accommodate more than one cable and wherein the second movable cover is configured to accommodate more than one cable.
9. A wire connector comprising: a base having a first end portion, a second end portion and a central portion, the base having an upper wall and a lower wall; a cover having a main cover member, a first movable cover member movably connected to a first side of the main cover member and a second movable cover member movably connected to a second side of the main cover member, the cover being attached to the base such that the main cover member is associated with the central portion of the base, the first movable cover member is associated with the first end portion of the base and the second movable cover member is associated with the second end portion of the base, the main cover member of the cover including a plurality of contact chambers, each contact chamber of the main cover member including at least one electrical contact positioned within a cavity of the contact chamber, a first passage to the first end portion of the base and a second passage to the second end portion of the base; a first cover coupling assembly associated with the first end portion of the base and the first movable cover member of the cover and used to releasably couple the first movable cover member of the cover to the first end portion of the base; and a second cover coupling assembly associated with the second end portion of the base and the second movable cover member of the cover and used to releasably couple the second movable cover member of the cover to the first end portion of the base.
10. The wire connector according to claim 9, wherein the main cover member further comprises at least one aperture configured to provide access to the electrical contact within the contact chamber.
11. The wire conductor of claim 10, wherein the at least one aperture further comprises a knockout configured to be removed to allow access therethrough.
12. The wire conductor of claim 10, wherein the at least one aperture further comprises a plug configured to be removed to allow access therethrough.
13. The wire connector of claim 10, further comprising a supplemental wire connector having a body and a movable cover movably connected to the body, the body including at least one aperture configured to receive a conductor; a supplemental wire connector coupling assembly associated with the body and the movable cover member of the supplemental wire connector and used to releasably couple the movable cover member of the supplemental wire connector to the body; and wherein the supplemental wire connector is configured to couple with the main cover member so that the apertures of the main cover align with the apertures of the supplemental wire connector.
14. The wire connector of claim 13, wherein the at least one aperture houses a connector contact configured to receive and retain a conductor so that the conductor is electrically connected with the connector contact.
15. The wire connector of claim 14, wherein the connector contact is at least one of a pin-type and a blade type connector contact.
16. The wire connector of claim 9, wherein the first movable cover is configured to accommodate more than one cable and wherein the second movable cover is configured to accommodate more than one cable.